

*"So starting from the top — everything is triggered by a GitHub Actions cron job that fires every weekday at 3:45 PM IST, right after market close. That single trigger kicks off two pipelines that run in parallel, not sequentially — you can see they branch off the same cron node.*

*On the left is the ML pipeline. It starts by pulling ten years of OHLCV — open, high, low, close, volume — price data for each stock, and engineers about 40 technical indicator features from that, things like RSI, MACD, ATR. Those features feed into three parallel model types for each of the 50 NIFTY stocks: a LightGBM classifier trained per stock, an HMM regime detector that figures out whether the overall market is in a bull, bear, or sideways state, and Chronos, a zero-shot forecasting model that predicts price trajectory without needing per-stock training. Those three outputs get blended — 60% LightGBM, 30% Chronos, 10% regime — into one ML signal.*

*On the right is the news pipeline — this is the LangGraph side, seven agents total. Agent 1 harvests articles from six RSS feeds and NewsAPI, with deduplication so the same story doesn't get processed twice. Agent 2 is a four-stage filter — cheapest checks first: a keyword blocklist to cut obvious noise, then named entity recognition to confirm the article even mentions a market-relevant entity, then FinBERT for financial sentiment scoring, and only the survivors go through the most expensive step, an LLM gate. If nothing survives that filter, the pipeline skips straight to Agent 5 — no point running mapping or scoring on zero articles. If articles do pass, Agent 3 maps them to specific NIFTY tickers, Agent 4 scores how severe the impact is, and Agent 5 cross-checks against live prices and validates whether yesterday's calls played out.*

*Both pipelines converge at Agent 6, the signal aggregator — it merges the ML signal and the news signal, roughly 60-40, and if they conflict with each other, it doesn't just pick one, it flags the conflict and lowers its own confidence. It also generates a plain-English explanation of its reasoning.*

*Agent 7 formats that into an alert, logs it to Supabase, and sends an email notification. From there, the Streamlit dashboard is stateless — it just reads fresh from Supabase on every page load, so there's no separate caching layer to keep in sync."*